

Induction and recursion (归纳和递归)

Chapter 5

Mathematical Induction (数学归纳法)

Section 5.1

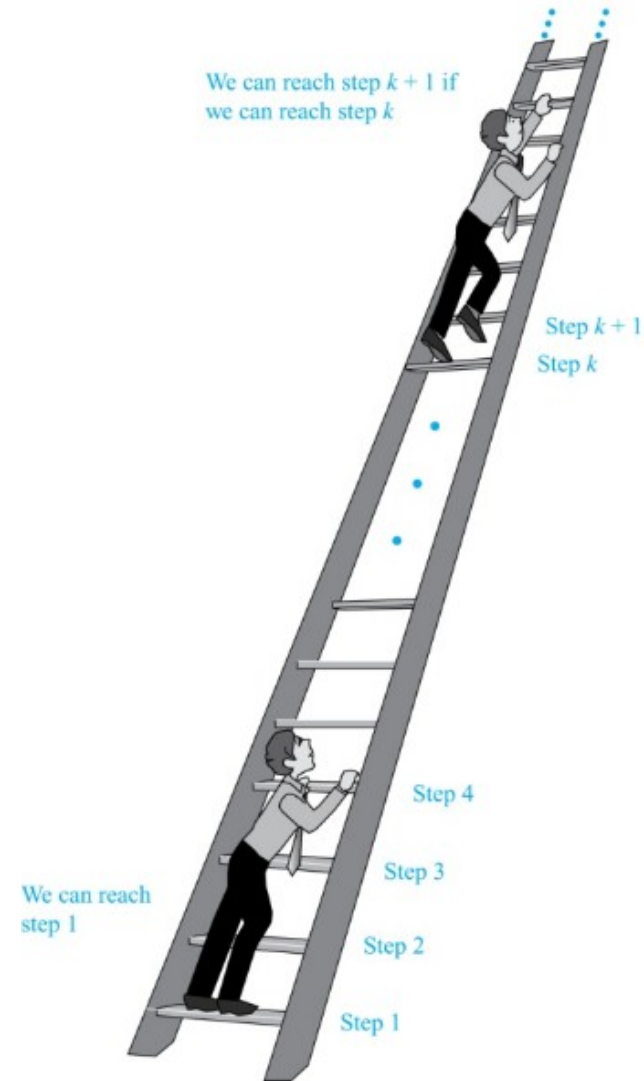
5.1.1 Climbing an Infinite Ladder

Suppose we have an infinite ladder:

1. We can reach the first rung(横档) of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

From (1), we can reach the first rung. Then by applying (2), we can reach the second rung. Applying (2) again, the third rung. And so on. We can apply (2) any number of times to reach any particular rung, no matter how high up.

This example motivates proof by **mathematical induction**(数学归纳法).



[Jump to long description](#)

5.1.2 Principle of Mathematical Induction

Principle of Mathematical Induction: To prove that $P(n)$ is true for all positive integers n , we complete these steps:

- *Basis Step*: Show that $P(1)$ is true.
- *Inductive Step* (归纳步): Show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step, assuming the *inductive hypothesis* that $P(k)$ holds for an arbitrary positive integer k , and show that under this assumption $P(k + 1)$ must also be true.

Important Points About Using Mathematical Induction

Mathematical induction can be expressed as the rule of inference

$$\left(P(1) \wedge \forall k (P(k) \rightarrow P(k+1)) \right) \rightarrow \forall n P(n),$$

where the domain is the set of positive integers.

In a proof by mathematical induction, we don't assume that $P(k)$ is true for all positive integers! We show that **if we assume that $P(k)$ is true, then $P(k+1)$ must also be true.**

Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer. We will see examples of this soon.

5.1.7 Examples of Proofs by Mathematical Induction

Proving Summation Formula

Example 2: Conjecture (推测) and prove correct a formula for the sum of the first n positive odd integers(奇数). Then prove your conjecture.

Solution: We have: $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, $1 + 3 + 5 + 7 + 9 = 25$.

- We can conjecture that the sum of the first n positive odd integers is n^2 ,
$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$
- We prove the conjecture is proved correct with mathematical induction.
- **BASIS STEP:** $P(1)$ is true since $1^2 = 1$.
- **INDUCTIVE STEP:** $P(k) \rightarrow P(k + 1)$ for every positive integer k .

Assume the inductive hypothesis holds and then show that $P(k + 1)$ holds as well.

Inductive Hypothesis: $1 + 3 + 5 + \cdots + (2k - 1) = k^2$

- So, assuming $P(k)$, it follows that:

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= [1 + 3 + 5 + \cdots + (2k - 1)] + (2k + 1) \\ &= k^2 + (2k + 1) \text{ (by the inductive hypothesis)} \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

- Hence, we have shown that $P(k + 1)$ follows from $P(k)$. Therefore the sum of the first n positive odd integers is n^2 .

Proving Inequalities₂

Example 6: Use mathematical induction to prove that $2^n < n!$, for every integer $n \geq 4$.

Solution: Let $P(n)$ be the proposition that $2^n < n!$.

- **BASIS STEP:** $P(4)$ is true since $2^4 = 16 < 4! = 24$.
- **INDUCTIVE STEP:** Assume $P(k)$ holds, i.e., $2^k < k!$ for an arbitrary integer $k \geq 4$. To show that $P(k + 1)$ holds:

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ &< 2 \cdot k! && \text{(by the inductive hypothesis)} \\ &< (k+1)k! \\ &= (k+1)! \end{aligned}$$

Therefore, $2^n < n!$ holds, for every integer $n \geq 4$.

Note that here the basis step is $P(4)$, since $P(0)$, $P(1)$, $P(2)$, and $P(3)$ are all false.

Proving Divisibility Results

Example: Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n .

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

- **BASIS STEP:** $P(1)$ is true since $1^3 - 1 = 0$, which is divisible by 3.
- **INDUCTIVE STEP:** Assume $P(k)$ holds, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k . To show that $P(k + 1)$ follows:

$$\begin{aligned}(k + 1)^3 - (k + 1) &= (k^3 + 3k^2 + 3k + 1) - (k + 1) \\ &= (k^3 - k) + 3(k^2 + k)\end{aligned}$$

By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. So by part (i) of Theorem 1 in Section 4.1, $(k + 1)^3 - (k + 1)$ is divisible by 3.

Therefore, $n^3 - n$ is divisible by 3, for every integer positive integer n .

Example 14: Tiling Checkerboards₁

Example: Show that every $2^n \times 2^n$ checkerboard(棋盘) with one square removed can be tile(平铺) using right triominoes(三格板).

A right triomino is an L-shaped tile which covers three squares at a time.



Solution: Let $P(n)$ be the proposition that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using right triominoes. Use mathematical induction to prove that $P(n)$ is true for all positive integers n .

- **BASIS STEP:** $P(1)$ is true, because each of the four 2×2 checkerboards with one square removed can be tiled using one right triomino.

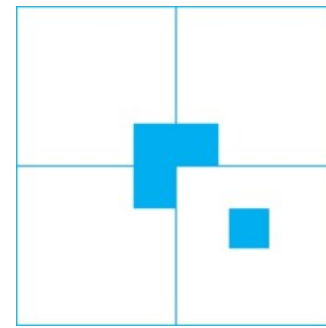
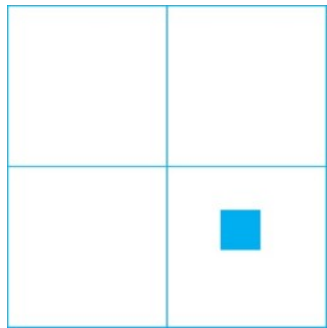


- **INDUCTIVE STEP:** Assume that $P(k)$ is true for every $2^k \times 2^k$ checkerboard, for some positive integer k .

Tiling Checkerboards₂

Inductive Hypothesis: Every $2^k \times 2^k$ checkerboard, for some positive integer k , with one square removed can be tiled using right triominoes.

Consider a $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed. Split this checkerboard into four checkerboards of size $2^k \times 2^k$, by dividing it in half in both directions.



Remove a square from one of the four $2^k \times 2^k$ checkerboards. By the inductive hypothesis, this board can be tiled. Also by the inductive hypothesis, the other three boards can be tiled with the square from the corner of the center of the original board removed. We can then cover the three adjacent squares with a triomino.

Hence, the entire $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed can be tiled using right triominoes.

- 9. a)** Find a formula for the sum of the first n even positive integers.
- b)** Prove the formula that you conjectured in part (a).

20. Prove that $3^n < n!$ if n is an integer greater than 6.

20. The basis step is $n = 7$, and indeed $3^7 < 7!$, since $2187 < 5040$. Assume the statement for k . Then $3^{k+1} = 3 \cdot 3^k < (k+1) \cdot 3^k < (k+1) \cdot k! = (k+1)!$, the statement for $k+1$.

Section Summary₁

Mathematical Induction

Guidelines for Proofs by Mathematical Induction

Examples of Proof by Mathematical Induction

Homework

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28. Find $a \text{ div } m$ and $a \text{ mod } m$ when

a) $a = -111, m = 99.$

$$-111 \text{ div } 99 = -2 \quad -111 \text{ mod } 99 = 87$$

b) $a = -9999, m = 101.$

$$-111 = 99 * (-2) + 87$$

$$-9999 \text{ div } 101 = 99 \quad -9999 \text{ mod } 101 = 0$$

$$-9999 = 101 * (99)$$

35. Decide whether each of these integers is congruent to 5 modulo 17.

a) 80

$$17 \nmid 80 - 5 \quad 5 \not\equiv 80 \pmod{17}$$

c) -29

$$17 \mid (-29) - 5 \quad 5 \equiv -29 \pmod{17}$$

Strong Induction and Well-Ordering

Section 5.2

Section Summary₂

Strong Induction

Example Proofs using Strong Induction

Using Strong Induction in Computational Geometry (*not yet included in overheads*)

Well-Ordering Property

Strong Induction

Strong Induction: To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, complete two steps:

- *Basis Step:* Verify that the proposition $P(1)$ is true.

- *Inductive Step:* Show the conditional statement

$$\left[P(1) \wedge P(2) \wedge \cdots \wedge P(k) \right] \rightarrow P(k+1)$$

holds for all positive integers k .

Strong Induction is sometimes called the *second principle of mathematical induction* or *complete induction*.

Strong Induction and the Infinite Ladder

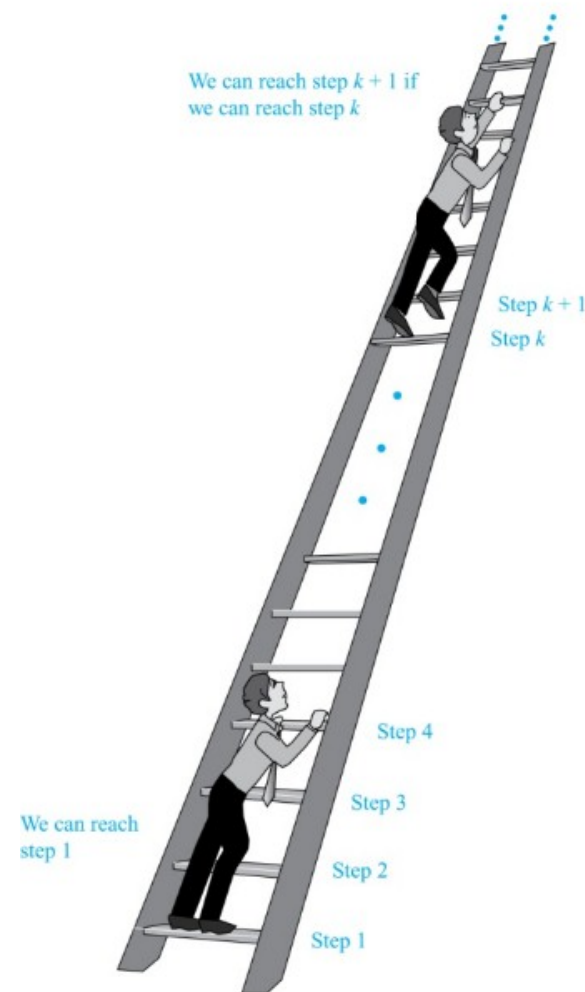
Strong induction tells us that we can reach all rungs if:

1. We can reach the first rung of the ladder.
2. For every integer k , if we can reach the first k rungs, then we can reach the $(k + 1)$ st rung.

To conclude that we can reach every rung by strong induction:

- BASIS STEP: $P(1)$ holds
- INDUCTIVE STEP: Assume $P(1) \wedge P(2) \wedge \dots \wedge P(k)$ holds for an arbitrary integer k , and show that $P(k + 1)$ must also hold.

We will have then shown by strong induction that for every positive integer n , $P(n)$ holds, i.e., we can reach the n th rung of the ladder.



Proof using Strong Induction₁

Example: Suppose we can reach the first and second rungs of an infinite ladder, and we know that if we can reach a rung, then we can reach two rungs higher. Prove that we can reach every rung. (Try this with mathematical induction.)

Solution: Prove the result using strong induction.

- **BASIS STEP:** We can reach the first step.
- **INDUCTIVE STEP:** The inductive hypothesis is that we can reach the first k rungs, for any $k \geq 2$. We can reach the $(k + 1)$ st rung since we can reach the $(k - 1)$ st rung by the inductive hypothesis.
- Hence, we can reach all rungs of the ladder.

Which Form of Induction Should Be Used?

We can always use strong induction instead of mathematical induction. But there is no reason to use it if it is simpler to use mathematical induction. (*See page 335 of text.*)

In fact, the principles of mathematical induction, strong induction, and the well-ordering property are all equivalent. (*Exercises 41-43*)

Sometimes it is clear how to proceed using one of the three methods, but not the other two.

Completion of the proof of the Fundamental Theorem of Arithmetic

Example: Show that if n is an integer greater than 1, then n can be written as the product of primes.

Solution: Let $P(n)$ be the proposition that n can be written as a product of primes.

- BASIS STEP: $P(2)$ is true since 2 itself is prime.
- INDUCTIVE STEP: The inductive hypothesis is $P(j)$ is true for all integers j with $2 \leq j \leq k$. To show that $P(k + 1)$ must be true under this assumption, two cases need to be considered:
 - If $k + 1$ is prime, then $P(k + 1)$ is true.
 - Otherwise, $k + 1$ is composite and can be written as the product of two positive integers a and b with $2 \leq a \leq b < k + 1$. By the inductive hypothesis a and b can be written as the product of primes and therefore $k + 1$ can also be written as the product of those primes.

Hence, it has been shown that every integer greater than 1 can be written as the product of primes.

(uniqueness proved in Section 4.3)

Proof using Strong Induction₂

Example: Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

Solution: Let $P(n)$ be the proposition that postage of n cents can be formed using 4-cent and 5-cent stamps.

- BASIS STEP: $P(12)$, $P(13)$, $P(14)$, and $P(15)$ hold.
 - $P(12)$ uses three 4-cent stamps.
 - $P(13)$ uses two 4-cent stamps and one 5-cent stamp.
 - $P(14)$ uses one 4-cent stamp and two 5-cent stamps.
 - $P(15)$ uses three 5-cent stamps.
- INDUCTIVE STEP: The inductive hypothesis states that $P(j)$ holds for $12 \leq j \leq k$, where $k \geq 15$. Assuming the inductive hypothesis, it can be shown that $P(k + 1)$ holds.
- Using the inductive hypothesis, $P(k - 3)$ holds since $k - 3 \geq 12$. To form postage of $k + 1$ cents, add a 4-cent stamp to the postage for $k - 3$ cents. Hence, $P(n)$ holds for all $n \geq 12$.

Proof of Same Example using Mathematical Induction

Example: Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

Solution: Let $P(n)$ be the proposition that postage of n cents can be formed using 4-cent and 5-cent stamps.

- BASIS STEP: Postage of 12 cents can be formed using three 4-cent stamps.
- INDUCTIVE STEP: The inductive hypothesis $P(k)$ for any positive integer k is that postage of k cents can be formed using 4-cent and 5-cent stamps. To show $P(k + 1)$ where $k \geq 12$, we consider two cases:
 - If at least one 4-cent stamp has been used, then a 4-cent stamp can be replaced with a 5-cent stamp to yield a total of $k + 1$ cents.
 - Otherwise, no 4-cent stamp have been used and at least three 5-cent stamps were used. Three 5-cent stamps can be replaced by four 4-cent stamps to yield a total of $k + 1$ cents.

Hence, $P(n)$ holds for all $n \geq 12$.

Well-Ordering Property₁

Well-ordering property 良序 性質 : Every nonempty set of nonnegative integers has a least element.

The well-ordering property is one of the axioms of the positive integers listed in Appendix 1.

The well-ordering property can be used directly in proofs, as the next example illustrates.

The well-ordering property can be generalized.

- **Definition:** A set is *well ordered* if every subset has a least element.
 - $\mathbf{N}$ is well ordered under $\leq$.
 - The set of finite strings over an alphabet using lexicographic ordering is well ordered.

Homework page362 3

Recursive Definitions and Structural Induction

Section 5.3

Section Summary₃

Recursively Defined Functions

Recursively Defined Sets and Structures

Structural Induction

Generalized Induction

Recursively Defined Functions₁

Definition: A recursive or inductive definition 递归或归纳定义 of a function consists of two steps.

- BASIS STEP: Specify the value of the function at zero.
- RECURSIVE STEP: Give a rule for finding its value at an integer from its values at smaller integers.

A function $f(n)$ is the same as a sequence $a_0, a_1, \dots$, where a_i , where $f(i) = a_i$. This was done using recurrence relations in Section 2.4.

Recursively Defined Functions₂

Example: Suppose f is defined by:

$$f(0) = 3,$$

$$f(n+1) = 2f(n) + 3$$

Find $f(1), f(2), f(3), f(4)$

Solution:

$$f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$$

$$f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$$

$$f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$$

$$f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$$

Example: Give a recursive definition of the factorial function $n!$:

Solution: $f(0) = 1$

$$f(n+1) = (n+1) \cdot f(n)$$

Fibonacci Numbers₁



Fibonacci
(1170- 1250)

Example : The Fibonacci numbers are defined as follows:

$$f_0 = 0$$

$$f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2}$$

Find f_2, f_3, f_4, f_5 .

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5$$

In Chapter 8, we will use the Fibonacci numbers to model population growth of rabbits. This was an application described by Fibonacci himself.

Next, we use strong induction to prove a result about the Fibonacci numbers.